

Atharva Bodade

Information Technology Undergraduate

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PROFESSIONAL SUMMARY

Information Technology Undergraduate focused on backend systems, low-level architecture, and local AI workflows. Hands-on experience developing RESTful services with FastAPI, integrating local language models via the Model Context Protocol (MCP), and exploring kernel-level development. Driven by a deep curiosity to understand software from the operating system layer up to distributed systems, actively documenting technical learnings online, and applying structured critical thinking to build reliability into software.

EDUCATION

Shri Sant Gajanan Maharaj College of Engineering

Bachelor of Engineering in Information Technology

CGPA: 7.14/10.0

Shegaon, Maharashtra, India

Expected May 2027

Relevant Coursework: Operating Systems (Kernel Dev), Data Structures & Algorithms, Computer Networks, Assembly Language Programming, Computer Organization and Architecture, Compiler Design.

Shri R.L.T. College of Science

Higher Secondary Certificate (HSC), General Science

MHT-CET: 94.43 Percentile · **Activities:** Distance Running

Akola, Maharashtra, India

Aug 2021 – May 2023

TECHNICAL SKILLS

Languages & Core: Python (AsyncIO), TypeScript, C, Assembly (x86_64), Bash, SQL, LaTeX.

Backend & Systems: FastAPI, Django, REST APIs, Microservices, System Design, Redis (Pub/Sub & Caching), PostgreSQL, Clean Architecture.

AI Tools & Frameworks: Model Context Protocol (MCP), Google ADK, LLM Integration, RAG (Pinecone, ChromaDB, LanceDB), Ollama.

DevOps & Environment: AWS (S3, Lambda), Docker, Kubernetes, Helm, CI/CD Fundamentals, Linux (Arch, Red Hat).

Frontend & Web: Next.js, React, Tauri, PyTorch, TensorFlow.

EXPERIENCE

VCF Cyber Solutions Pvt. Ltd.

December 2025 – March 2026

Cyber Security Intern (Remote)

- Analyzed security architecture across **7 core infrastructure domains** (Network, Cloud, Endpoint, IAM) to identify common systemic vulnerabilities and operational attack vectors.
- Performed hands-on vulnerability assessments using **Nmap**, **Wireshark**, and **Aircrack-ng** across **15+ subnet segments**, flagging **10+ network misconfigurations**.
- Replicated attack vectors using **Metasploit** and authored **20+ technical reports** mapping exploit patterns (e.g., API key exposure, MFA fatigue) to practical mitigations.

Apexa IQ

September 2025 – October 2025

Software Engineer Intern (Remote)

- Built a **Python REST API (FastAPI)** to bridge core backend business logic with user interfaces, maintaining sub-500ms latency across high-throughput endpoints.
- Authored **40+ pages of detailed technical documentation** (github.com/UnsettledAverage73/apexaiq-docs) for internal tools, reducing developer setup time by **2 weeks**.
- Containerized microservice environments using **Docker**, standardizing deployment workflows across development and local test instances.

Mozilla Open Source Community

August 2023 – August 2024

Technical Advisor

- Led hands-on technical sessions on **Git**, **Linux command-line tools**, and **Web Security** for **50+ junior students**, increasing active open-source participation.

- Mentored **15+ peers** on open-source contribution practices and code review standards, leading to **5 accepted Pull Requests** in external community projects.

KEY PROJECTS

- **Average IDE (Local Development Environment)** *TypeScript, Python, MCP, SLMs*
 Link: github.com/UnsettledAverage73/Locally-AI-Integrated-IDE
 Built an autonomous, privacy-focused IDE designed to run language models locally without sending proprietary source code to third-party APIs.
 - **Native MCP Integration:** Implemented the **Model Context Protocol** to connect local language models directly to workspace tools (File System, Terminal, Web Search) for local ingestion.
 - **Execution Loop:** Implemented a "ReAct" execution flow allowing local models to analyze directory trees, propose project structures, and run local build scripts autonomously.
 - **Context Management:** Designed context windowing logic to pass raw program state to quantized models (Ollama) for reliable multi-file refactoring on local hardware.
- **Zoho OS (x86_64 Bare-Metal Kernel)** *C, Assembly (NASM), QEMU, Multiboot2*
 Link: github.com/UnsettledAverage73/zoho-os
 Written a zero-dependency 64-bit operating system kernel from scratch to explore low-level memory management, hardware interrupts, and system mechanics.
 - **Bootstrapping & Long Mode:** Wrote a multi-stage boot sequence transitioning execution from 16-bit Real Mode to 64-bit Long Mode with identity-mapped 4-level paging (PML4).
 - **Memory & Multiprocessing:** Implemented an $O(1)$ physical frame allocator, virtual memory manager (VMM), and a round-robin scheduler with per-CPU queues for up to 16 cores.
 - **File System & Framebuffer:** Implemented a Virtual File System (Initrd/Tar), hardware drivers (Intel E1000, XHCI), and a framebuffer drawing engine optimized via Dirty Rectangle redrawing.
- **Predictive Maintenance AI System** *Python, FastAPI, Next.js, Redis, Pinecone, Ollama, Docker*
 Link: github.com/UnsettledAverage73/predictive_maintainance_system
 Developed a real-time monitoring tool for industrial IoT telemetry streams to detect sensor anomalies and suggest automated maintenance prescriptions.
 - **Local Inference Pipeline:** Configured local LLM execution (Ollama) with fallback handling, combining Sarvam AI for multilingual log parsing and Pinecone for historical context.
 - **Event Pipeline:** Handled live telemetry streams using **Redis Pub/Sub** messaging, passing updates directly to a real-time **Next.js** frontend dashboard.
 - **Automated Alerts:** Configured real-time notification triggers via WhatsApp API (Twilio) and asynchronous event handlers for instant threshold alerts.
- **CIS Benchmark Audit Tool (SIH 2024 Finalist)** *Python, Bash, Shell, React, OS Security*
 Link: github.com/UnsettledAverage73/his-2k24
 Developed an automated cross-platform auditor for the Smart India Hackathon (NTRO problem statement) to evaluate system configurations against CIS security baselines.
 - **Multi-OS Hardening Engine:** Wrote automated inspection scripts for enterprise Linux (RHEL 8/9, Ubuntu 20.04/22.04) and Windows environments to detect 20+ critical misconfigurations.
 - **Reporting Engine:** Built desktop (Tkinter) and web (React) visualizers paired with an automated engine generating JSON exports and executive PDF audit reports.

HONORS & COMMUNITY

- **Finalist, Smart India Hackathon 2024 (NTRO Problem Statement):** Selected among top national teams for building an automated multi-OS system audit tool evaluating Center for Internet Security (CIS) benchmarks.
- **Second Runner-up, Navonmesh Hackathon (SSGMCE):** Built an industrial IoT predictive maintenance tool evaluated around Tata Technologies requirements for real-time anomaly detection.
- **Top 15 Finalist, Woxsen University Hackathon (2026):** Presented **Average IDE** among 150+ teams; built a feature to provision and manage AWS EC2 instances directly through local IDE commands.
- **Participant, Matrix AI Protocol Hackathon:** Built a risk forecasting model for peer-to-peer lending using XGBoost regression and classification algorithms.
- **Certifications:** AWS Academy Graduate (Cloud Security), Google Cloud Networking Fundamentals.